QUESTION BANK

📘 MODULE 1 – ETHICS & GENDER STUDIES (Question Bank)

◆ Short Answer (2–3 marks)

1. Define ethics.
2. Differentiate between personal ethics and professional ethics.
3. What is civic virtue?
4. Define professionalism.
5. What is plagiarism?
6. What is cyber trust?
7. Define confidentiality.
8. What is gender identity?
9. Define gender stereotypes.
10. What is gender spectrum?

◆ Medium Answer (4–6 marks)

11. Explain the importance of integrity in design and research.
12. Discuss the role of diligence and responsibility in engineering.
13. Explain the concept of data, information, and knowledge.
14. Describe challenges in balancing law and ethics.
15. Explain gender disparity in education and employment.
16. Discuss the role of confidentiality in moral integrity.
17. Explain the impact of digital revolution on ethics.
18. Describe gender expression and sexuality.

◆ Long Answer (8–10 marks)

19. Explain professional ethics with examples.
20. Discuss plagiarism and its consequences in academics and research.
21. Explain codes of ethics and their importance in engineering practice.
22. Discuss gender discrimination in everyday life with examples.
23. Explain women's contributions to Science & Technology.

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24. Discuss gender equity, diversity, and justice.

🌱 MODULE 2 – ENVIRONMENTAL ETHICS & ECOSYSTEMS

🔹 Short Answer

1. Define environmental ethics.
2. What is anthropocentrism?
3. Define biocentrism.
4. What is ecocentrism?
5. Define biodiversity.
6. What is an ecosystem?
7. What is sustainable engineering?
8. Define triple bottom line.
9. What is life cycle analysis?
10. Define urban ecology.

🔹 Medium Answer

11. Explain the importance of environmental ethics.
12. Compare anthropocentrism and ecocentrism.
13. Explain triple bottom line with examples.
14. Discuss biodiversity conservation.
15. Explain human impact on ecosystems.
16. Describe sustainable urban planning.
17. Explain landscape ecology principles.
18. Discuss ecosystems in Kerala/India.

🔹 Long Answer

19. Explain key philosophical theories of environmental ethics.
20. Discuss sustainable engineering principles in detail.
21. Explain biodiversity loss and conservation methods.

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22. Discuss urbanization and its environmental impacts.
23. Explain green infrastructure and its benefits.
24. Analyse a case of ecosystem degradation.

💧 MODULE 3 – WATER, WASTE & TRANSPORTATION

◆ Short Answer

1. Define hydrology.
2. What is water cycle?
3. Define water scarcity.
4. What is zero waste?
5. Define recycling.
6. What is circular economy?
7. Define degrowth.
8. What is sustainable mobility?
9. What is E-mobility?
10. Define environmental flow.

◆ Medium Answer

11. Explain water management practices.
12. Discuss causes of water pollution.
13. Explain principles of zero waste.
14. Compare linear and circular economy.
15. Explain degrowth principles.
16. Discuss sustainable transportation design.
17. Explain integrated mobility systems.
18. Discuss waste reduction strategies.

◆ Long Answer

19. Explain hydrology and water cycle in detail.

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20. Discuss water scarcity and its solutions.
21. Explain circular economy with examples.
22. Discuss zero waste initiatives (case study).
23. Explain environmental impacts of transportation.
24. Discuss sustainable mobility solutions in India.

⚡ MODULE 4 – ENERGY, CLIMATE & POLICIES

🔹 Short Answer

1. Define renewable energy.
2. List types of renewable energy sources.
3. What is climate change?
4. Define carbon footprint.
5. What is mitigation?
6. What is adaptation?
7. Define environmental policy.
8. What is sustainability?
9. Define resilience.
10. What is biomass energy?

🔹 Medium Answer

11. Explain solar and wind energy.
12. Discuss challenges in renewable energy adoption.
13. Explain impacts of climate change.
14. Discuss climate change in India/Kerala.
15. Explain role of engineers in sustainability.
16. Discuss environmental regulations.
17. Explain sustainable technologies.
18. Discuss policy implementation challenges.

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- **Long Answer**

19. Explain renewable energy sources in detail.
20. Discuss engineering solutions to climate change.
21. Explain environmental policies (national & international).
22. Analyse a real-world sustainability case study.
23. Discuss future trends in environmental sustainability.
24. Explain the role of engineers in building a sustainable future.